Digvijay Singh

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EDUCATION

Last Updated on 6/2026

- **Ph.D. - Johns Hopkins University School of Medicine** 8/2012 - 1/2018
- **Integrated BS-MS - Indian Institute of Technology (IIT), Kharagpur** 7/2007 - 4/2012

PROFESSIONAL EXPERIENCES

Staff Scientist at Singular Genomics 9/2024 –
- Develop assays and tools; spanning image-processing, bioinformatics, & AI for a multi-omics platform (G4x)

National Institute of Health-K99 fellow at University of California (UC), San Diego 2/2023 – 9/2024
- Co-led a team to solve structure-model of Nuclear Basket; helped develop AI-driven bio-tools (4 manuscripts).

Damon Runyon fellow at UC San Diego 2/2019 – 2/2023
- Co-led/worked in teams to solve structures of Nuclear-Pore & condensates & co-authored a book on tomography.

Postdoctoral & Visible Molecular Cell fellow at UC San Diego 2/2018 – 2/2019
- Revitalized a project detailing the steps of in-cell structural biology culminating in a highly cited publication.

PhD researcher at Johns Hopkins University & UIUC 6/2012 – 1/2018
- Co-led/worked in teams (including with Prof. Doudna, a Nobelist) to advance CRISPR/T4 & sm-microscopy tools.

Multiple internships at top-tier universities & companies (listed below), all awarded with stipends
- University of Cambridge [6/2012–8/2012], IIT [8/2011-3/2012], MIT [5/2011-7/2011], KIT-Germany [11/2010–12/2010].
- Univ. of Illinois (UC) [5/2010–7/2010], Unilever [11/2009–12/2009], General Electric [4/2009–7/2009].
- Performed computational modeling, protein purification, kinetic analysis, & organic synthesis across diverse projects.

SKILLS

- **Computational Biology, Biophysics & Structural biology**: Bioinformatics & image processing, range of light-electron microscopy & structural biology techniques/tools (e.g., Super-resolution, cryoEM, cryo-FIB).
- **Computing & AI**: Scientific computing (arrays, linear algebra, dataframes); deep learning (CUDA, PyTorch, Keras, Transformers, CNNs); data pipelines/orchestration (Snowflake, Airflow); HPC & parallel computing.
- **Cell Biology & Biochemistry**: Multi-omics assays, Cell culture, CRISPR, Cloning, PCR, mutagenesis, Protein purification, Gel-electrophoresis based assays, labeling.

SELECT PRESENTATIONS (Invited or Selected): 3DEM GRC (2023), Biophysical Society Meeting (2023, 2022, 2017, 2015), Southern California cryo-EM symposium (2022), Cryo-EM workshop at Brookhaven National Lab, USA (2022, 2023), National Tomography Workshop (3 lectures & workshops each year), AIIMS, New Delhi (2022-2026), Friends of Cell Meeting, San Diego Cluster of Institutes (2022), CRISPR Workshop at CSIR-IGIB, New Delhi (2019), Physics of Living Systems Conference (2015), Center for Physics of Living Cells Symposium, UIUC (2015).

EXPERT SERVICES, TEACHING & MENTORSHIP

- **Reviewer for:** *Nature Structural & Molecular Biology, Proceedings of the National Academy of Science, Nature Communications, Journal of Biological Chemistry, Chemical Science, Biochemistry, ACS Omega, Scientific Reports, Cellular & Molecular Life Sciences, Frontiers in Molecular Neuroscience, Frontiers in Neuroscience, JoVE.*
- **Instructor/Teaching assistant** at *Physics of Living Cells summer schools* & University of Illinois (2013-15).
- **Mentor for** 3 students (UC San Diego), 3 students (Johns Hopkins University)

PUBLICATIONS [23 so far; 9 first/co-first*] [Citations > 3000]

Selected:

19. **D. Singh***, N. Soni*, J. Hutchings*, .., E. Villa. The Molecular Architecture of the Nuclear Basket. *Cell* (2024).
16. C. Akey*, **D. Singh***, C. Ouch*, I. Echeverria*, ... E. Villa, M. Rout. Comprehensive structure & functional adaptations of the yeast nuclear pore complex. *Cell* (2022). Media-Release
4. **D. Singh**, ..., J. A. Doudna, T. Ha. Real-time observation of DNA recognition & rejection by the RNA-guided endonuclease Cas9. *Nature Communications* (2016).
8. **D. Singh**, ..., T. Ha. Real-time observation of DNA target interrogation & product release by .. Cas12a. *PNAS* (2018).
7. **D. Singh**, ..., T. Ha. Mechanism of improved specificity of engineered Cas9s revealed by single molecule analysis. *Nature Structural & Molecular Biology* (2018).
15. L. Dai*, **D. Singh***, ..., V. B. Rao. A viral genome packaging ring-ATPase is a flexibly coordinated pentamer. *Nature Communications* (2021).
9. I. Okafor*, **D. Singh***, Y. Wang*, .., T. Ha. Single molecule analysis of effects of non-canonical guide RNAs & specificity-enhancing mutations on Cas9-induced DNA unwinding. *Nucleic Acids Research* (2019).

18. **D. Singh**, E. Villa. Cryo-Focused Ion Beam Milling of Cells. *Springer book on Cryo-Electron Tomography* (2023).
6. **D. Singh**, T. Ha. Understanding the molecular mechanism of CRISPR toolbox using single-molecule approaches. *ACS Chemical Biology* (2018).
3. J. Fei, **D. Singh**,... , T. Ha. Single-molecule analysis of the target search kinetics of a bacterial sRNA. *Science* (2015).

Remaining:

13. H. Yu, .., **D. Singh**, ..TDP-43 & HSP70 phase separate into anisotropic, intranuclear liquid spherical annuli. *Science* (2020).
12. S. Lu, .., **D. Singh**, ..The SARS-CoV-2 Nucleocapsid phosphoprotein forms ...condensates with RNA & ..the M protein. *Nature Communi.* (2020).
20. Y. Liang, .., **D. Singh**, ..RobPicker: A Meta Learning Framework for Robust Identification of Macromolecules in Cryo-Electron Tomograms.
21. Y. Liang, ..., **D. Singh**, ..Multi-Modal Large Language Model Enables All-Purpose Prediction of Drug Mechanisms & Properties.
22. Li Zhang, .., **D. Singh**, ..A tri-modal contrastive learning framework for protein representation learning. *Cell Reports* (2026).
11. Y. Wang, .., **D. Singh**, ..Real-time observation of Cas9 postcatalytic domain motions. *PNAS* (2020).
10. F. Wagner, .., **D. Singh**, ..Preparing samples from whole cells using focused-ion-beam milling for cryo-electron tomography. *Nature Protocols* (2020).
1. A. Dhar, .., **D. Singh**,..Different protein stability & folding kinetics in nucleus .. & cytoplasm of living cells. *Biophysical Journal* (2011).
2. B. Hua, .., **D. Singh**, ..An improved surface passivation method for single-molecule Studies. *Nature Methods* (2014).
17. B. Raveh, .., **D. Singh**, ..Integrative spatiotemporal map of nucleocytoplasmic transport. *PNAS* (2025).
23. A. Abadia, .., **D. Singh**, ..TissuStamp: A high-throughput tissue transfer workflow for FFPE spatial biology . *Journal of Histotechnology* (2026).
5. B. Hua, .., **D. Singh**, ..Single-molecule centroid localization algorithm improves the accuracy of fluorescence binding assays. *Biochemistry* (2018).
14. A. Poddar, .., **D. Singh**, ..Effects of individual base-pairs on in vivo target search & destruction kinetics of small RNA. *Nature Communi.* (2021).

.. → *Please refer to the link for complete author list.*

AWARDS & FUNDING

- NIH K99-R00 Pathway to Independence Award, National Institute of Health. 2023 - 2028
- Damon Runyon Fellowship, Damon Runyon Cancer Foundation 2019 - 2023
- Virtual Molecular Cell Consortium (VMCC) Fellow, UC San Diego. 2018 - 2019
- Finalist of Damon Runyon-Dale F. Frey Award for Breakthrough Scientists 2022
- Biophysical Society Education Travel Award 2017
- Finalist of International Howard Hughes Medical Institute fellowship 2015
- Johns Hopkins Biophysics department nominee for international Weintraub Award 2018
- INSPIRE fellowship by Government of India 2008 - 2012